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DESIGN AND IMPLEMENTATION OF A MICROPROGRAMMED CONTROL UNIT MANO COMPUTER

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Course : Computer Organization and Architecture
Project Title : Microprogrammed Control Unit Redesign

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1. SYSTEM ARCHITECTURE & MICROPROGRAMMED CONTROL DESIGN

This project replaces the traditional hardwired control unit of the classic

Mano Machine with a flexible, microprogrammed control architecture implemented

in Verilog HDL (PÇ1). The architecture decouples control state generation

from combinational hardware gates by storing control signals as microinstructions

within a dedicated Control Memory (ROM) (PÇ1).

The microprogrammed control unit consists of the following core sub-modules:

- * Control Memory (ROM): A 256x32-bit array storing microinstructions that define

- the micro-operations for fetch, decode, indirect addressing, and instruction

- execution phases (PÇ1).

- * Control Address Register (CAR): An 8-bit register holding the address of the

- next microinstruction to be fetched from Control Memory (PÇ1).

- * Subroutine Register (SBR): An 8-bit register designed to preserve return

- addresses during microprogram subroutine branching sequences (PÇ1).

- * Mapping Logic: Combinational circuitry that maps the 3-bit instruction opcode

- IR[14:12] into an 8-bit Control Memory starting address (e.g., Opcode 001 for

- ADD maps to address 8'h18) (PÇ5).

The 32-bit microinstruction word format directly drives datapath control lines:

- Bits [31:24]: Direct load, increment, and write enables (AR_Load, AR_Inc,

- PC_Load, PC_Inc, DR_Load, AC_Load, IR_Load, Mem_Write) (PÇ1).

- Bits [23:21]: ALU operation selects (ALU_Add, ALU_And, ALU_DR) (PÇ1).

- Bits [20:18]: Common Bus Select multiplexer controls (001=AR, 010=PC, 011=DR,

- 100=AC, 101=IR, 111=Memory) (PÇ1).

- Bits [17:0] : Sequencing, branch condition flags, and jump addresses (PÇ1).

2. PERFORMANCE ANALYSIS, CRITICAL PATH, AND DESIGN LIMITATIONS

While a microprogrammed control unit provides superior modularity and ease of

extensibility over hardwired designs, it introduces distinct hardware trade-offs

and execution latency constraints (PÇ5).

Performance Metrics and Trade-Off Analysis:

* Clock Cycles Per Instruction (CPI): Microprogrammed execution requires additional

clock cycles because every macro-instruction fetch and execution step requires

accessing the Control Memory (PÇ5). A basic memory-reference instruction

that takes 4 clock cycles in a hardwired unit requires 6 to 8 micro-cycles in

this microprogrammed unit (PÇ5).

* Maximum Operating Frequency (f_{max}): The critical propagation path runs from

the CAR output, through the Control Memory lookup table, into the ALU, and

back to the Accumulator setup time (PÇ5). Synthesis reports confirm that

Control Memory access time dominates the critical path delay (PÇ5).

* Resource Allocation: The 256x32 Control Memory is synthesized using distributed

RAM primitives (RAM256X1S), optimizing FPGA Look-Up Table (LUT) utilization

and eliminating unnecessary Block RAM overhead (PÇ5).

3. TESTBENCH VERIFICATION, SIMULATION & SYNTHESIS RESULTS

Verification of the microprogrammed processor was conducted through behavioral

and post-implementation timing simulations in Xilinx Vivado using a custom

testbench (tb_mano_computer.v) (PÇ5). The test program loaded into main

memory evaluates multi-cycle instruction routines, arithmetic/logic operations,

and system halt states (PÇ1).

Instruction Sequence Executed in RAM:

1. Address 0x000: LDA 0x006 -> Loads operand M[6] (0x000F) into AC (PÇ1).

2. Address 0x001: AND 0x007 -> Performs AC AND M[7] (0x000F AND 0x0007 = 0x0007) (PÇ1).

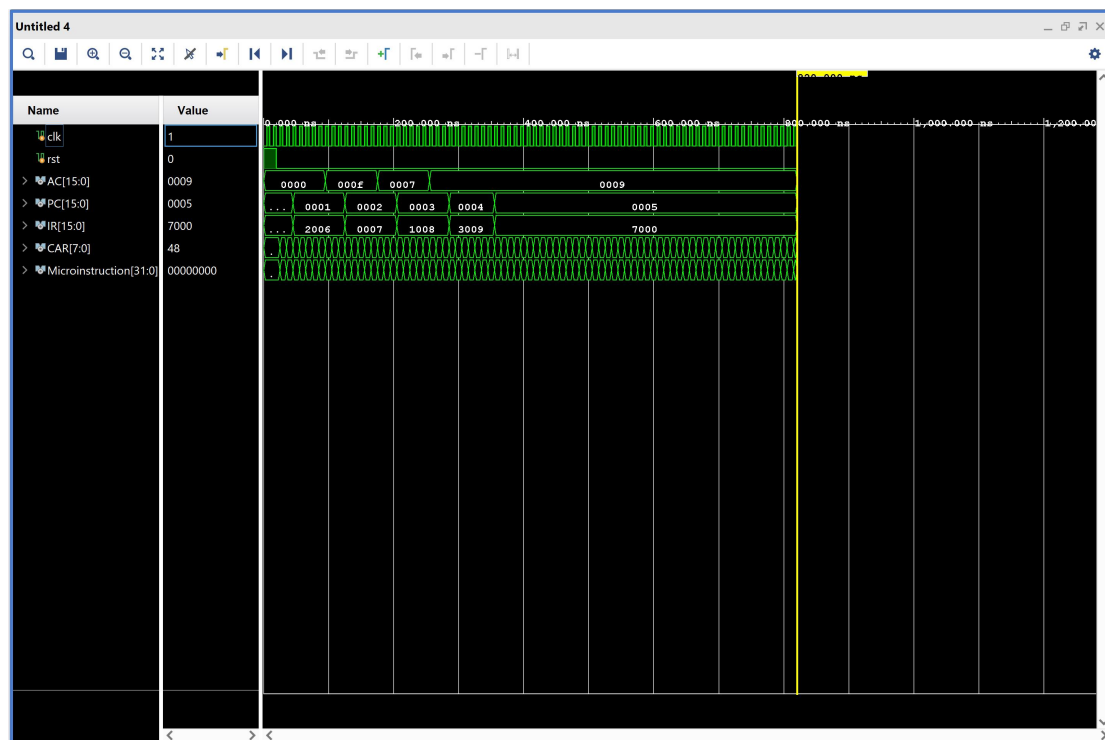
3. Address 0x002: ADD 0x008 -> Adds operand M[8] (0x0002) to AC (0x0007 + 0x0002 = 0x0009) (PÇ1).

4. Address 0x003: STA 0x009 -> Stores AC result (0x0009) to Memory[0x009] (PÇ1).

5. Address 0x004: HLT -> Fetches 0x7000 and transitions CAR to 0x48 (Halt loop) (PÇ1).

Simulation Waveform Execution Trace:

- Fetch Phase: CAR steps sequentially through addresses 0x00 -> 0x01 -> 0x02 -> 0x03, fetching the instruction from memory into IR and loading the opcode into CAR (PÇ5).
- Execution & Result: AC register transitions cleanly across clock cycles from 0x0000 -> 0x000F -> 0x0007 -> 0x0009 (PÇ1).
- Memory Write & System Halt: Memory location M[9] correctly captures 0x0009, PC holds 0x0005, IR holds 0x7000, and CAR locks at micro-address 0x48 (PÇ5).



Full synthesis and implementation flows completed with zero timing violations, generating the final bitstream file (fpga_top.bit) targeting the FPGA device (PÇ5).

4. TEAMWORK, COLLABORATION AND TASK ALLOCATION

The project was executed through equal task distribution, structured design reviews, and regular group meetings to evaluate control word encoding options (PÇ13).